

Outline

Transdisciplinary approach

Computational Thinking, Analysis & Problem Solving

Technology-Enabled Education

Connecting Concepts to Everyday Life

Developing Sensitivity and Scientific Temper

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Connecting concepts across disciplines to create a more coherent and meaningful learning experience.

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Example: Understanding the Human Ear

The human ear is usually introduced as a topic in **Biology**.

In a **Physics** chapter on Sound, the ear can be introduced to help students understand how sound is detected and heard.

This connects the physics of sound with the biology of hearing, showing students that scientific ideas do not exist in isolated subjects.

Transdisciplinary approach

SOUND 17

How Do We Hear and Process Sound

The human ear converts sound waves into electrical signals for the brain. First, the outer part of our ear, called the pinna, collects sound waves and guides them down the ear canal. At the end of this canal, the sound waves strike a thin membrane called the eardrum, making it vibrate like the skin of a drum. The eardrum is connected to three tiny bones inside the ear. These bones are called the malleus, incus, and stapes. These bones amplify the vibrations. The amplified vibrations pass into the cochlea, a snail-shaped tube filled with liquid. As the liquid inside the cochlea ripples, it stimulates tiny hair cells inside the cochlea. These hair cells convert mechanical vibrations into electrical signals. The auditory nerve carries electrical signals to the brain, which interprets them as speech or music.

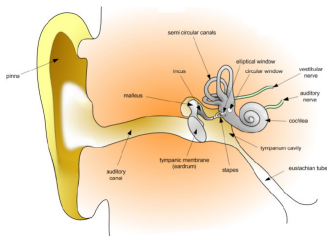


Figure 26: Diagram of the auditory system of the human ear. Image source: SASOL Inzalo Foundation.

Computational Thinking, Analysis & Problem Solving

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Example: Understanding Frequency

SOUND 13

EXERCISE: Look at the wave diagram in Figure 22, it shows a sound wave recorded over a time period of 2 seconds. Mark and label the regions of compressions and rarefactions on the diagram.

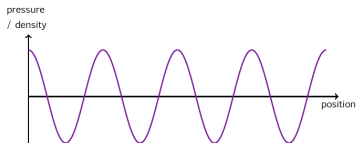


Figure 22: Sound wave recorded over a time period of 2 seconds.

EXERCISE: Count the total number of complete wave cycles that occur within the 2-second time period shown in the figure. A cycle (or wave cycle) is one complete pattern of a wave. For a sound wave, one single cycle consists of one full compression plus one full rarefaction (or the distance from the start of one compression to the start of the next). It represents one complete back-and-forth vibration.

EXERCISE: Use your count from the previous exercise to calculate the frequency of the wave. Frequency is the number of complete cycles per second, measured in Hertz (Hz).

Technology-Enabled Education

Using digital tools to extend what students can observe, measure, and investigate.

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Example: Investigating Pitch

Students first explore pitch through the Singing Pipe and Glass activity.

With the help of a smartphone and the **Phyphox** app, the teacher can measure the frequency of the sounds produced.

Students can then compare their observations with the measured frequencies.

Technology-Enabled Education

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SOUND 14

Pitch

Sounds with higher frequency have a higher pitch and sounds with lower frequency have a lower pitch. A mosquito makes a high-pitched sound. A drum makes a low-pitched sound. Which one is vibrating faster?

Activity: Singing Pipe and Glass

Materials required

1. A glass, transparent cup, or water bottle
2. A thin hollow pipe
3. water

Procedure

Fill the glass about three-quarters full with water. Place the pipe vertically in the water leaving one end of the pipe above the water. Blow gently across the open end of the pipe. Listen to the sound produced. Next, move the pipe up so that a longer part of the pipe is above the water. Blow across the open end again. Move the pipe down so that a shorter part of the pipe is above the water. Blow across the open end again. Compare the sounds.

OBSERVE: Does the sound change when you change the length of the air column?

Ask your teacher to help you use the **Phyphox** app on their smartphone to measure the frequency of the sounds. Compare the frequency of the high-pitched and low-pitched sounds.

Connecting Concepts to Everyday Life

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Example: How Do We Form Different Sounds?

In the chapter on Sound, students can explore how the sounds they use in everyday speech are produced.

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SOUND 5

How Do We Form Different Sounds?

The sound produced by the vocal cords is only a basic buzzing sound. This sound is shaped into speech by different parts of our mouth. Our tongue, lips, teeth, the roof of the mouth (palate), and the nose work together to produce different sounds and words.

Letter	How the sound is made with your mouth, nose, and tongue?
ㅍ (pa)	Close both lips tightly, then open them suddenly
ㅁ (ma)	Close both lips. Let the sound come out through your nose
ㅌ (tha)	Touch the tip of your tongue to your upper front teeth.
ㄴ (la)	Touch the tip of your tongue to the bumpy part just behind your upper front teeth.
ㄷ (la)	Curl the tip of your tongue slightly backwards and touch the roof of your mouth.
ㄹ (zha)	Curl the tip of your tongue slightly backwards, but do not let it touch the roof of your mouth.

Developing Sensitivity and Scientific Temper

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Example: Noise Pollution & Hearing Loss

After learning about sound, students are also taught about noise pollution, and how it affects people and other living beings.

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SOUND 20

Noise Pollution

Unwanted or unpleasant sound is called noise. When there is too much noise in our surroundings, it leads to noise pollution.

Common sources of noise pollution include traffic noise from vehicles and horns, loudspeakers used at high volume, industrial noise from machines and factories, construction activities.

Continuous exposure to loud noise can cause stress, lack of sleep, and hearing problems. How can we reduce noise pollution?

We can reduce noise pollution by avoiding unnecessary honking on the roads, keeping the volume of loudspeakers low, using silencers and maintaining vehicles properly, planting trees, which can help absorb sound, following rules and regulations about noise levels.

Reducing noise pollution helps create a healthier and more peaceful environment.

Hearing loss and hearing aids

Sounds above 85 dB (such as loud headphones, factory machines, or heavy traffic) can bend and snap the delicate hair cells inside the cochlea. Broken hair cells cannot regrow. Hearing aids are devices that help by amplifying incoming sound waves so remaining hair cells can detect them.

Image of hearing aid

Figure 28: Image of hearing aid